

Buzzard cluster mock data vector with optical-selection effects: forward model, analytical and empirical $\mathcal{B}_{\text{sel}}$ closures

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Abstract

This document is the technical reference for `MockDataVector.ipynb`: a Buzzard-halo DES Y3-like cluster mock that layers the Costanzi+2026 (C26) optical selection systematics on top of the raw simulation outputs, on the DES Y1 cluster binning ($\lambda \in \{20, 30, 45, 60, \infty\}$, $z \in \{0.20, 0.35, 0.50, 0.65\}$). The data vector is constructed along *two parallel routes* that share the same Costanzi+2019 (C19) forward model upstream and the same lensing geometry: an **analytical** closure based on the C26 Appendix C Eq. C1 fitting function applied to the C19-binned stack (cheap enough for an MCMC chain), and an **empirical** ratio that stacks on the catalog redMaPPer outputs (`LAMBDA_CHISQ`, `Z_LAMBDA`) and divides by a $(\log M, z^{\text{tr}})$ -matched random reference (Wu+2022, Costanzi+2026). The notebook also produces a binned number-count vector $N(\lambda^{\text{ob}}, z^{\text{ob}})$. Boost-factor contamination is not included (deferred work). All outputs are saved to `output/MockDataVector.npz`; the headline plots embedded in this document live in `output/figs/`.

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1 Pipeline topology

The notebook executes the following stages in order:

```
halo_load → photoz_proxy → sample_lambda_true → sample_lambda_obs → lensing_inputs
→ number_counts → analytical_C1 → empirical_mass_match → save_npz
```

The first four stages are the C19 forward model: select host halos above $\log_{10} M_{\text{vir}} = 13$ in $z \in [0.20, 0.65]$ (with the Buzzard $0.33 \leq z < 0.37$ box seam excised), add a DES Y3-like Gaussian photo- z scatter to produce z^{ob} , and draw λ^{tr} and λ^{ob} from the C26/C19 kernels. `lensing_inputs` centralises the per-halo profiles and $\Sigma_{\text{crit}}^{-1}$ table, plus the catalog redMaPPer columns (`LAMBDA_CHISQ`, `Z_LAMBDA`) used by the empirical route. The two lensing-data-vector stages are independent and produce their outputs on the same DES Y1 binning so the user can compare them side-by-side. The final stage persists everything to `output/MockDataVector.npz`.

2 Shared conventions

Cosmology. FlatLambdaCDM($H_0=70$, $\Omega_m=0.3$, $\Omega_b=0.046$, $T_{\text{cmb}}=2.725$) via `astropy.cosmology`, matching the Buzzard simulation cosmology used by the upstream halo catalogue.

Units. Per-halo Σ , $\Delta\Sigma$ are stored in physical $M_{\odot}/\text{pc}^2$; radii `rp_phys_mpc` (from `src/radial_bins_phys_mpc.py`) are in physical Mpc, with $N_R = 15$ log-spaced bins between 0.0323 and 30 Mpc. Halo masses M_{vir} are in $h^{-1} M_{\odot}$.

Halo selection. Mirrors 0-MakeMock.ipynb: `pid == -1` (host halos only), $0 \leq \cos i \leq 1$ (measured), drop the $0.33 \leq z < 0.37$ simulation-box seam, then require $0.20 \leq z^{\text{tr}} \leq 0.65$ and $\log_{10} M_{\text{vir}} \geq 13$. Yields $\sim 5.83 \times 10^5$ halos.

DES Y1 binning (used everywhere). Both abundance and lensing bins are the DES Y1 cluster edges (Costanzi+2019, McClintock+2019):

$$\text{LBDBINS} = \{20, 30, 45, 60, 500\}, \quad \text{ZMIN_LIST} = \{0.20, 0.35, 0.50\}, \quad \text{ZMAX_LIST} = \{0.35, 0.50, 0.65\}.$$

The 500 caps the open $[60, \infty)$ richness range. There are $N_{\lambda} \times N_z = 4 \times 3 = 12$ bins.

Photo- z proxy. Per-halo Gaussian

$$z^{\text{ob}} = z^{\text{tr}} + \sigma_z(1 + z^{\text{tr}})\mathcal{N}(0, 1), \quad \sigma_z = 0.01,$$

clipped at $z^{\text{ob}} \geq 0.02$. Stand-in for the DES Y3 redshift kernel of Myles+2021; captures the dominant scatter but not the tails.

Random seeds. A single `numpy.random.default_rng(seed=42)` is shared across the forward model. The MC-vs-PDF sanity histogram uses `seed=123`; the §8.1 bootstrap uses `seed=0`.

3 The intrinsic mass–richness relation (C26 Eq. 15)

Path `src/costanzi_selection.py (l_sat, l_tr, sample_lambda_true)`. The mean satellite richness follows C26 Eq. 15,

$$\langle \lambda_{\text{sat}} | M, z \rangle = \left(\frac{M - M_{\text{min}}}{M_{\text{pivot}}} \right)^\alpha \left(\frac{1 + z}{1 + z_{\text{piv}}} \right)^\epsilon, \quad (1)$$

clipped to zero below M_{min} . The mean intrinsic richness is $\langle \lambda^{\text{tr}} | M, z \rangle = 1 + \langle \lambda_{\text{sat}} | M, z \rangle$ for $M \geq M_{\text{min}}$ and zero below (C26 Sec. III).

The DES Y1 NC+3×2pt best-fit constants, hard-coded in `src/costanzi_selection.py`:

$$\log_{10} M_{\text{min}} = 11.385, \quad \alpha = 0.859, \quad \log_{10} M_1 = 12.696, \quad \sigma_{\text{intr}} = 0.181, \quad \epsilon = 0.284, \quad z_{\text{piv}} = 0.4544,$$

with $M_{\text{pivot}} = M_1 - M_{\text{min}}$.

The per-halo λ^{tr} is drawn from a compound Poisson with multiplicative- lognormal broadening of the Poisson mean (C26 Sec. III):

$$m_{\text{broad}} = m_{\text{sat}} \exp\left[\sigma_{\text{intr}} \mathcal{N}(0, 1) - \frac{1}{2} \sigma_{\text{intr}}^2\right], \quad N_{\text{sat}} \sim \text{Poisson}(m_{\text{broad}}), \quad \lambda^{\text{tr}} = \mathbb{I}[M \geq M_{\text{min}}] + N_{\text{sat}}.$$

4 The projection kernel (C19 Eq. 6)

Path `src/costanzi_selection.py (projection_params, sample_lambda_obs)`. C19 Eq. 6 is a Gaussian-core + exponentially-modified-Gaussian (EMG) mixture:

$$P(\lambda^{\text{ob}} | \lambda^{\text{tr}}, z) = (1 - f_{\text{prj}}) \mathcal{N}(\mu, \sigma) + f_{\text{prj}} \text{EMG}(\mu, \sigma, \tau). \quad (2)$$

The four parameters $\{\mu, \sigma, \tau, f_{\text{prj}}\}$ depend on λ^{tr} with z -dependent coefficients calibrated from SDSS redMaPPer (C19 Appendix A, Eq. A8):

$$\begin{aligned} \mu(\lambda^{\text{tr}}, z) &= a_\mu(z) + b_\mu(z) \lambda^{\text{tr}}, & \sigma(\lambda^{\text{tr}}, z) &= b_\sigma(z) (\lambda^{\text{tr}})^{a_\sigma(z)}, \\ \tau(\lambda^{\text{tr}}, z) &= b_\tau(z) (\lambda^{\text{tr}})^{-a_\tau(z)}, & f_{\text{prj}}(\lambda^{\text{tr}}, z) &= b_{f_{\text{prj}}}(z) (1 + e^{-\lambda^{\text{tr}}})^{-a_{f_{\text{prj}}}(z)}. \end{aligned} \quad (3)$$

The 15 rows of `data/prj_params_DESY3_lss_lin_dep_getdist_v1.txt` are interpreted as posterior samples and collapsed to the posterior mean via `load_prj_posterior_mean`.

The mixture is sampled directly using the $\text{EMG} = \mathcal{N}(\mu, \sigma) + \text{Exp}(1/\tau)$ representation, avoiding a closed-form EMG density evaluation in the sampling path. The closed form is used only for the sanity overlay below.

Sanity check 1 — MC draws vs analytical kernel. Histogram repeated `sample_lambda_obs` draws at fixed (λ^{tr}, z) and overlay the closed-form C19 PDF; they should agree within Poisson noise.

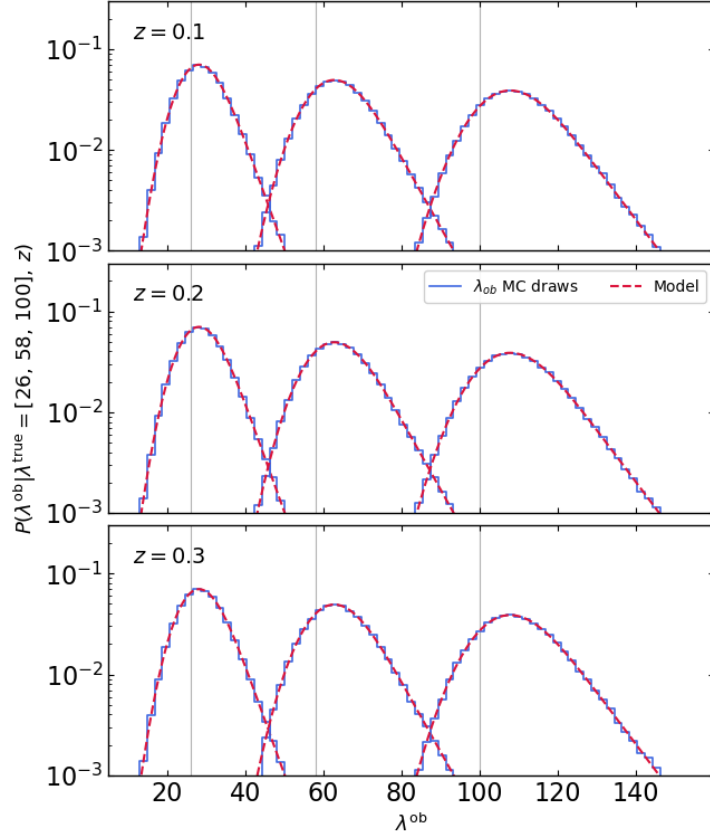


Figure 1: MC draws of λ^{ob} from `sample_lambda_obs` (blue histograms) overlaid with the closed-form PDF of Eq. 2 (red dashed) at three reference redshifts and three intrinsic richnesses. The two agree within Poisson noise across the bulk and the tail.

Sanity check 2 — Gaussian core + EMG tail decomposition. Same kernel as Eq. 2, split into the two physical components. At large λ^{tr} the EMG tail dominates the high- λ^{ob} side and drives the projection bias; this is what the analytical $\mathcal{B}_{\text{sel}, \text{C1}}$ of §6 fits.

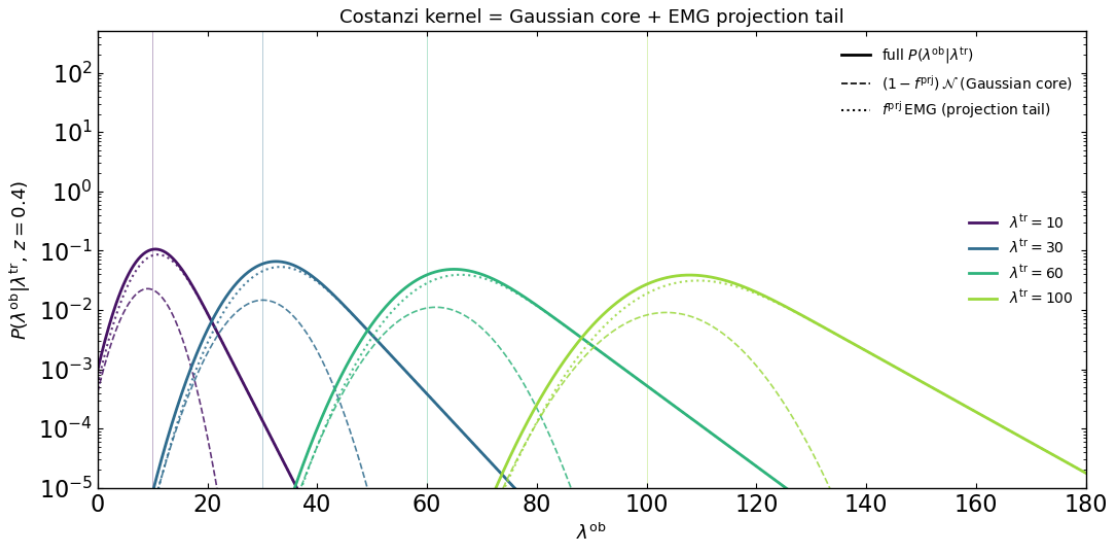


Figure 2: Decomposition of the kernel of Eq. 2 into its $(1 - f_{\text{prj}})\mathcal{N}$ Gaussian core (dashed) and f_{prj} EMG projection tail (dotted) at $z = 0.4$ for four intrinsic richnesses $\lambda^{\text{tr}} \in \{10, 30, 60, 100\}$. The EMG tail rises with λ^{tr} and drives the optical-selection bias on the lensing data vector.

5 Output 1 — $N(\lambda^{\text{ob}}, z^{\text{ob}})$ number counts

We obtain $N(\lambda^{\text{ob}}, z^{\text{ob}})$ by counting the mock halos that fall into each $(\lambda^{\text{ob}}, z^{\text{ob}})$ bin on the DES Y1 grid. Both axes are forward-model outputs rather than catalog redMaPPer columns: the C19 sampler depends only on (M, z) and carries no per-halo LSS information, which matches the inputs of the analytical $\mathcal{B}_{\text{sel}, \text{C1}}(\lambda^{\text{ob}}, z^{\text{ob}})$ of §6 and keeps the abundance and lensing sides of the data vector in the same C26/C19 model world. $N(\lambda^{\text{ob}}, z^{\text{ob}})$ is therefore the correct entry for a C26 likelihood integrating Eq. 1 into Eq. 2.

6 Lensing data vector — ANALYTICAL route (Matteo / C26 Eq. C1)

Binning quantity. The C19 forward-model $(\lambda^{\text{ob}}, z^{\text{ob}})$. Each λ^{ob} is an independent draw from the kernel of Eq. 2 conditioned only on (λ^{tr}, z) : the kernel encodes the LSS-driven projection statistics through its EMG tail at the population level, but per-halo λ^{ob} values are not correlated with the actual line-of-sight structure around each halo and therefore carry no information correlated with the per-halo lensing signal.

Selection bias. redMaPPer counts red-sequence galaxies inside a projected aperture with no line-of-sight resolution, so projected LSS preferentially boosts λ^{RM} for halos in over-dense filaments and lifts the stacked $\Delta\Sigma(R)$ above the mass-only expectation. C26 Appendix C captures this lift with a 5-parameter fitting function (no integration); Matteo (priv. comm.) reports the same form fits the bias on $\Delta\Sigma$:

$$\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R) = A \left(\frac{R}{R_0} \right)^\alpha \left[1 + \left(\frac{R}{R_0} \right)^\gamma \right]^{(\beta-\alpha)/\gamma} + C, \quad (4)$$

with $(A, \alpha, \beta, \gamma, C) = (0.12, 4.11, -0.18, 1.82, 1.00)$, $R_0 = R_\lambda(\lambda^{\text{ob}})(1+z^{\text{ob}})$, and $R_\lambda = (\lambda^{\text{ob}}/100)^{0.2} h^{-1} \text{ Mpc}$ (C26 Appendix B).

The C26 paper writes the published form for Σ with a fixed +1 rather than a free C ($C = 1$ limit of Eq. 4); we plot both flavours for comparison. The data vector is

$$\Delta\Sigma_{\text{obs}}(R) = \mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R) \Delta\Sigma_{\text{stack}}(R), \quad \langle \gamma_t(R) \rangle = \Delta\Sigma_{\text{obs}}(R) \Sigma_{\text{crit}}^{-1} \left(\langle z^{\text{ob}} \rangle_{\text{bin}} \right),$$

with $\Delta\Sigma_{\text{stack}}$ the unweighted mean of `DeltaSigma` across the C19-binned halos in each $(\lambda^{\text{ob}}, z^{\text{ob}})$ cell.

Why a fitting function? An MCMC chain evaluates the lensing model $\sim 10^4$ – 10^6 times, so the per-step cost matters. Computing the full Costanzi-2026 prediction from scratch requires nested integrals over halo mass and redshift at every step — prohibitive at chain length. Equation (4) replaces those integrals with five constants per bin, cutting the cost from seconds to microseconds without changing the answer.

Selection-bias ratio $\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R)$ The fitting function evaluated at the bin-mean $(\langle \lambda^{\text{ob}} \rangle, \langle z^{\text{ob}} \rangle)$ for every $(\lambda^{\text{ob}}, z^{\text{ob}})$ cell, with the published Σ -flavour overlay.

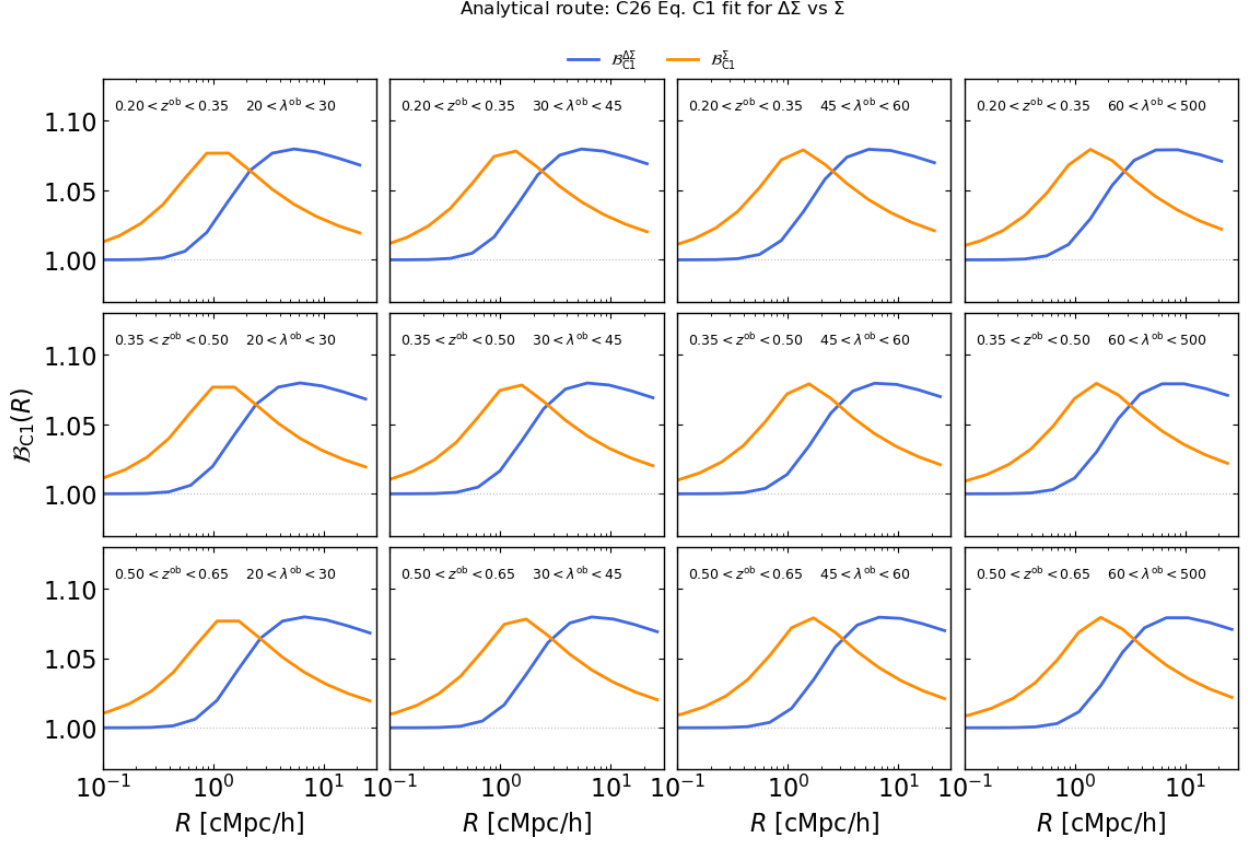


Figure 3: Analytical route: C26 Eq. C1 selection-bias ratio per $(\lambda^{\text{ob}}, z^{\text{ob}})$ bin on the DES Y1 grid. Blue: $\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}$ (used in the data vector). Orange: $\mathcal{B}_{\text{sel}, \text{C1}}^{\Sigma}$ overlay. x -axis is comoving R in h^{-1} Mpc to match the C26 convention; y -axis is the bias factor.

7 Lensing data vector — EMPIRICAL route (Heidi / mass-matched ratio)

Binning quantity. The catalog redMaPPer outputs $(\lambda^{\text{RM}}, z^{\text{RM}})$, i.e. LAMBDA_CHISQ and Z_LAMBDA. These come from running the redMaPPer cluster finder on the Buzzard photometric catalog and *do* carry per-halo LSS information.

Selection bias. Built directly from the catalog as the ratio of the redMaPPer-selected stack to a $(\log M, z^{\text{tr}})$ -matched random reference, following Wu+2022 method iii (`src/stacked_profile_weighted_by_mass_redshift`). For each $(\lambda^{\text{RM}}, z^{\text{RM}})$ bin:

$$\langle \Delta\Sigma(R) \rangle_{\text{rnd}} = \frac{\sum_c w_c \langle \Delta\Sigma(R) | c \rangle_{\text{full mock}}}{\sum_c w_c}, \quad w_c = N_c^{\text{sel}}, \quad (5)$$

summed over $(\log_{10} M, z^{\text{tr}})$ cells of width $(\Delta \log M, \Delta z) = (0.1, 0.05)$. The cell averages are taken over the *full* mock, so the reference resamples halos that share the selected sample’s mass and redshift but were never preferentially picked for their environment. The empirical bias is

$$\mathcal{B}_{\text{sel}}^{\text{emp}}(R) = \frac{\langle \Delta\Sigma(R) \rangle_{\text{sel}}}{\langle \Delta\Sigma(R) \rangle_{\text{rnd}}}.$$

No fitting function, no $R \rightarrow 0$ extrapolation, both numerator and denominator come straight from DeltaSigma.

Bootstrap. The reported $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ is the **bootstrap mean** over $N_{\text{boot}} = 50$ resamples of halos within each bin (with mass-matched reference rebuilt at every step), capturing the statistical uncertainty driven by finite halo count. The shaded bands in the panel grids are the bootstrap standard deviation; the saved file keeps only the bootstrap mean.

7.1 Selection-bias ratio $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ with bootstrap bands

We evaluate $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ on the redMaPPer-binned $(\lambda^{\text{RM}}, z^{\text{RM}})$ grid using Eq. (5) and report the bootstrap mean and 1σ band per bin. The expected signature is the bell-shaped excess peaking near $R \sim 1$ pMpc that Wu+2022 (Fig. 2) identify as the LSS-environment-driven selection bias: at smaller scales the one-halo term dominates and the ratio approaches unity, while at larger scales the projected over-density that boosted λ^{RM} falls outside the lensing aperture and the ratio relaxes back toward one.

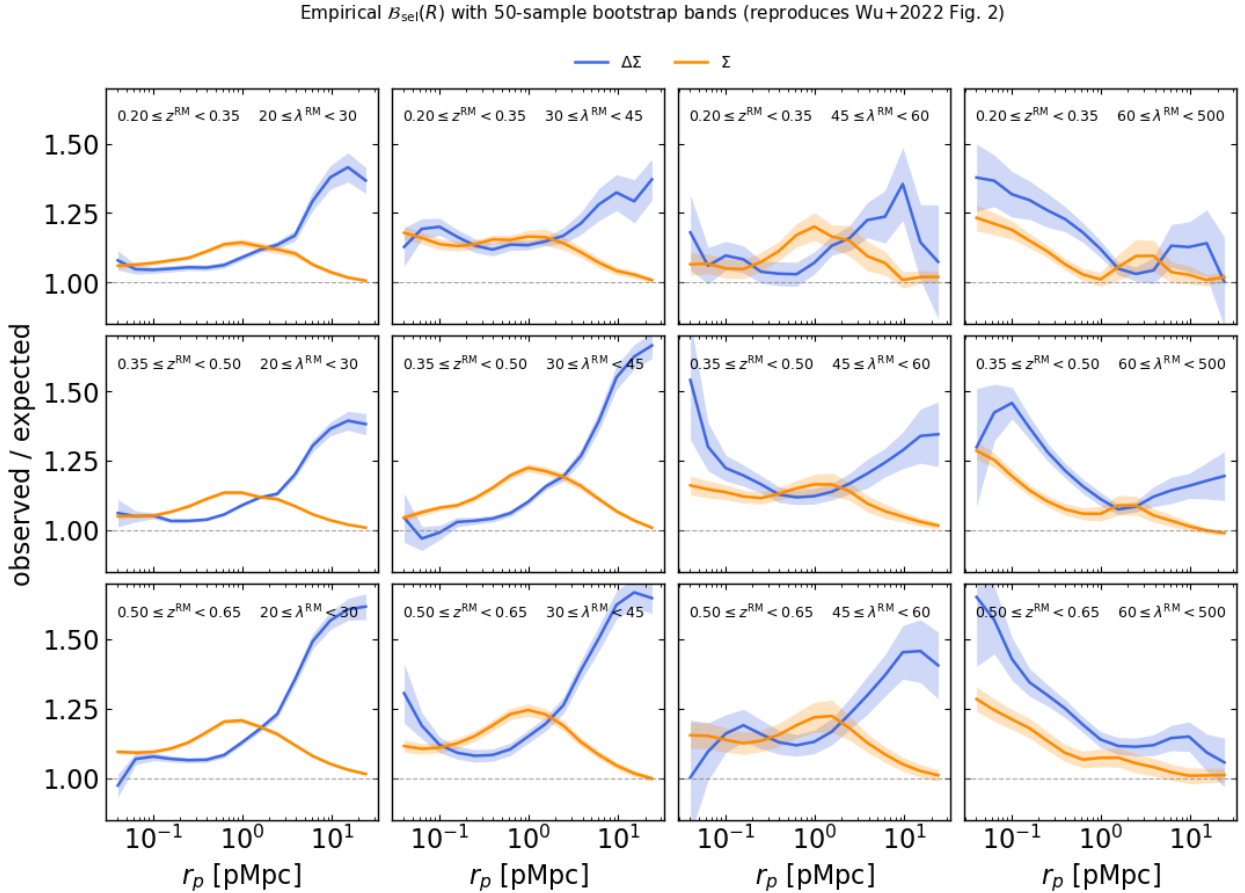


Figure 4: Empirical route: $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ per $(\lambda^{\text{RM}}, z^{\text{RM}})$ bin on the DES Y1 grid, with $N_{\text{boot}} = 50$ bootstrap bands. Blue: $\Delta\Sigma$ ratio. Orange: Σ ratio (overlay). Reproduces Wu+2022 Fig. 2: the bell-shaped peak around $R \sim 1$ pMpc is the LSS-environment-driven selection bias signal.

7.2 Null test: replace λ^{RM} with the C19 sampler

If we re-bin on the C19 forward-model $(\lambda^{\text{ob}}, z^{\text{ob}})$ instead of the catalog $(\lambda^{\text{RM}}, z^{\text{RM}})$, the same mass-matched-random pipeline should return unity at every R , because the C19 sampler is by construction blind to per-halo LSS. Any residual deviation from 1 is the bias of the matching procedure itself, not a physical signal.

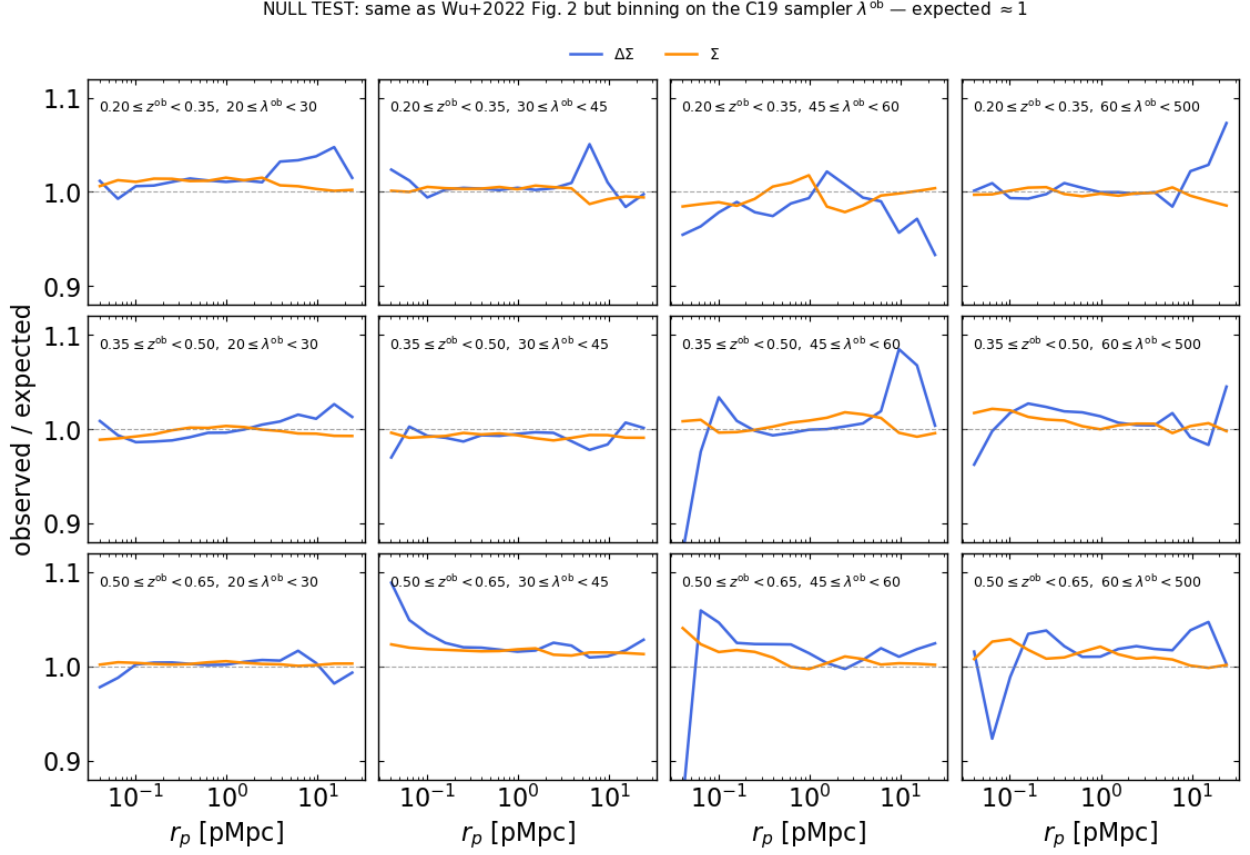


Figure 5: Null test: same Wu+22 panel grid but binning on the C19 sampler ($\lambda^{\text{ob}}, z^{\text{ob}}$) rather than the catalog ($\lambda^{\text{RM}}, z^{\text{RM}}$). The bell collapses to ≈ 1 across all bins, confirming the $R \sim 1$ pMpc peak in Fig. 4 is genuinely LSS-driven.

7.3 Tangential shear: redMaPPer-selected vs mass-matched random

The bias ratio of §7.1 is most directly seen by comparing the two tangential-shear stacks that enter its numerator and denominator. The mass-matched random reference is the γ_t a hypothetical unbiased selector would measure on halos of the same $(\log M, z^{\text{tr}})$ distribution; the redMaPPer-selected stack is the γ_t an actual cluster finder produces. Their offset is the observable counterpart of $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ and the quantity an MCMC must reproduce.

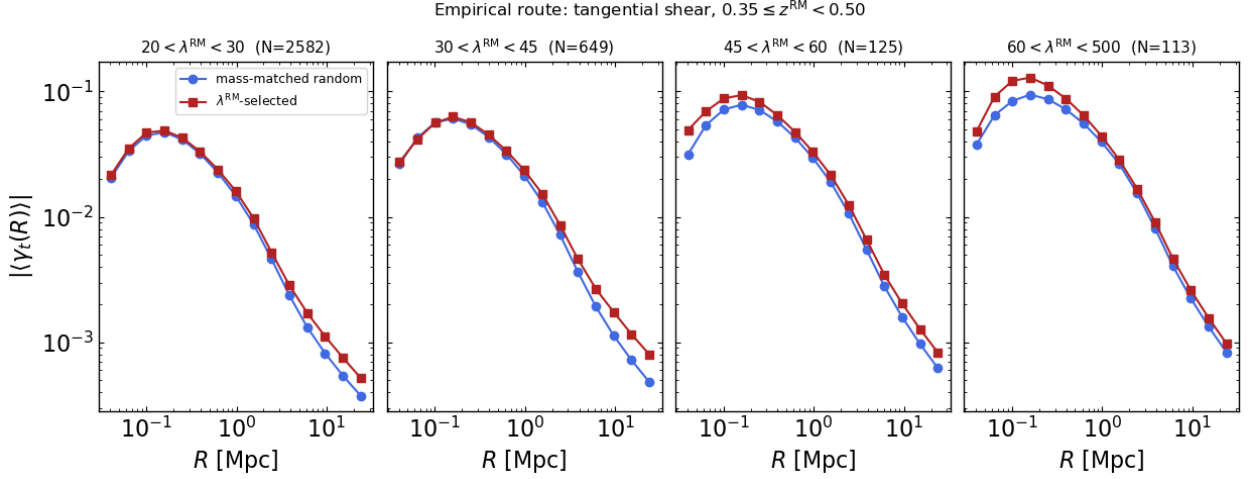


Figure 6: Empirical route: per-bin tangential shear in the middle redshift slice $z^{\text{RM}} \in [0.35, 0.50)$. Blue circles: γ_t from the mass-matched random reference. Red squares: γ_t from the λ^{RM} -selected stack (the observed signal). The selected curve sits above the reference by the $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ factor of Fig. 4.

8 Output archive

The final notebook cell writes both routes plus bin metadata to a single `output/MockDataVector.npz`. With $N_\lambda = 4$, $N_z = 3$, $N_R = 15$:

Key	Shape	Content
<i>Bin metadata</i>		
<code>radii_phys_mpc</code>	$(N_R,)$	15-bin physical radial grid (Mpc)
<code>lambda_bins</code>	$(N_\lambda + 1,)$	DES Y1 richness edges $\{20, 30, 45, 60, 500\}$
<code>z_bin_min</code>	$(N_z,)$	lower-redshift edges
<code>z_bin_max</code>	$(N_z,)$	upper-redshift edges
<i>Number counts (forward-model side)</i>		
<code>NC</code>	(N_λ, N_z)	$N(\lambda^{\text{ob}}, z^{\text{ob}})$
<i>Analytical route, C19-binned</i>		
<code>gamma_t_stack_C19</code>	(N_λ, N_z, N_R)	stacked γ_t , no correction
<code>gamma_t_obs_C1</code>	(N_λ, N_z, N_R)	stack $\times \mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}$
<code>B_sel_C1</code>	(N_λ, N_z, N_R)	analytical $\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R)$
<i>Empirical route, RM-binned</i>		
<code>gamma_t_stack_RM</code>	(N_λ, N_z, N_R)	$(\log M, z^{\text{tr}})$ -matched random reference γ_t
<code>gamma_t_obs_RM</code>	(N_λ, N_z, N_R)	λ^{RM} -selected stack γ_t (the observed signal)
<code>B_sel_emp</code>	(N_λ, N_z, N_R)	bootstrap-mean $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$

Table 1: Output-archive layout. Default shapes are $(4, 3)$ for the counts and $(4, 3, 15)$ for the radial vectors.

9 Known simplifications

- **Photo- z kernel.** The Gaussian $\sigma_z = 0.01(1 + z^{\text{tr}})$ proxy is a stand-in for the redshift- and colour-dependent kernel of Myles+2021. It captures the dominant scatter but misses the tails that drive the high- z outlier rate.

- **Boost-factor contamination** not modelled. Cluster-member contamination of the source sample (Y3 boost factors) is the other major Y3 lensing systematic; it stacks on top of the optical-selection bias treated here.
- **Posterior-mean projection coefficients.** The C19 projection parameters are evaluated at the posterior mean of the upstream chain. Row-by-row covariance propagation is not implemented.
- **Single Y3 realisation.** The §8.1 empirical ratio is computed on one Y3 realisation; Wu+2022 averages 12 Y1 + 1 Y3 realisations and is correspondingly tighter.

10 File map

- `MockDataVector.ipynb` — the production notebook. Top-to-bottom execution reproduces every figure and writes the output archive.
- `src/costanzi_selection.py` — C26 Eq. 15 sampler and C19 Eq. 6 mixture sampler, plus the legacy single-scale $\mathcal{B}_{\text{sel}}$ weight.
- `src/stacked_profile_weighted_by_mass_redshift.py` — Wu+22 / Esteves (M, z)-matched random reference builder.
- `src/radial_bins_phys_mpc.py` — the 15-bin physical radial grid `rp_phys_mpc`.
- `data/prj_params_DESY3_lss_lin_dep_getdist.v1.txt` — 15 posterior samples of the 10 C19 projection coefficients.
- `src/fileLoc.py` — machine-resolved path table for the raw Buzzard halo catalogue, profile output, and β_{eff} table.
- `output/MockDataVector.npz` — saved data vector (Tab. 1).
- `output/figs/` — headline plots embedded in this PDF.
- `docs/MockDataVector.tex` — this document.

11 References

- Costanzi et al. 2026, *Forward analytical model for the optical selection bias on galaxy cluster lensing profiles*. Eqs. 15 (mass–richness), Appendix C Eq. C1 (selection-bias fit), Sec. III (Poisson–Gaussian convolution).
- Costanzi et al. 2019, arXiv:1807.07072. Eq. 6 (Gaussian + EMG mixture), Appendix A Eq. A8 (per-parameter forms).
- Wu et al. 2022, arXiv:2203.05416. Fig. 2 (selection-bias panel grid) and Appendix B (mass–redshift weighted reference).
- McClintock et al. 2019, MNRAS 482, 1352 — DES Y1 cluster binning convention used throughout.